Detecting Manipulative Recommendation Systems Using Jacobian Spectral Distortion

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Abstract

Recommender systems are widely used in e-commerce platforms, streaming services, online marketplaces, and social media applications to provide personalized recommendations based on user behavior and interaction history. Although these systems improve user engagement and decision making, they are vulnerable to manipulation, hidden recommendation bias, fake rating amplification, and unfair promotional influence.

Traditional evaluation metrics such as accuracy, RMSE, precision, recall, and NDCG mainly evaluate recommendation quality but fail to detect hidden internal distortions within the recommendation engine. Manipulative systems can therefore continue producing highly accurate recommendations while still unfairly amplifying selected content.

This work proposes a Jacobian spectral distortion framework for detecting manipulation in recommender systems. The recommendation engine is mathematically modeled as a transformation from user input space to recommendation output space. Jacobian sensitivity analysis is used to study how small perturbations in user inputs affect recommendation outputs. Singular Value Decomposition (SVD) is then applied to identify dominant amplification directions and hidden spectral distortions.

A manipulation detection strategy based on abnormal singular value amplification is introduced to identify recommendation systems exhibiting excessive sensitivity toward suspicious input features. The framework provides a model-level manipulation detection mechanism unlike conventional user-behavior-based approaches.

Index Terms— recommender systems, Jacobian matrix, singular value decomposition, manipulation detection, spectral distortion, recommendation bias, sensitivity analysis, anomaly detection

1 INTRODUCTION

Recommender systems have become a fundamental component of modern digital platforms. Applications such as Netflix, YouTube, Amazon, Spotify, and Instagram rely heavily on recommendation algorithms to personalize content for users.

These systems analyze:

- user ratings
- watch history
- purchase history
- likes and clicks
- browsing patterns
- interaction behavior

Using this information, recommendation engines predict items that users are likely to prefer.

Despite their effectiveness, recommender systems are highly vulnerable to hidden manipulation. Fake ratings, sponsored promotions, injected interactions, and biased optimization objectives can artificially influence recommendation outputs.

Manipulation can produce several harmful effects:

- unfair promotion of products or movies
- suppression of competing content
- biased recommendation exposure
- reduction in recommendation fairness
- loss of user trust

Existing approaches mostly focus on external behavioral analysis such as:

- fake-user detection
- abnormal rating analysis
- temporal interaction monitoring
- behavioral anomaly detection

However, these methods rarely analyze the internal mathematical behavior of the recommendation model itself.

This project proposes a mathematical framework that studies internal recommendation sensitivity using Jacobian matrices and Singular Value Decomposition. The framework identifies hidden amplification behavior and abnormal sensitivity patterns that indicate possible manipulation.

2 PROBLEM STATEMENT

Modern recommender systems optimize primarily for engagement and recommendation accuracy. As a result, hidden manipulative behavior can remain undetected while standard evaluation metrics still report good performance.

Several major challenges exist:

- Small malicious perturbations can create disproportionately large recommendation changes.
- Fake ratings can amplify selected content unfairly.
- Existing metrics cannot reveal hidden internal instability.
- Current methods rely heavily on user behavior instead of model sensitivity.
- Internal recommendation amplification patterns remain largely unexplored.

Therefore, a mathematically grounded framework is required to detect hidden manipulation directly from the internal dynamics of recommendation systems.

3 OBJECTIVES

The primary objectives of this work are:

- Represent recommender systems as mathematical transformations.
- Analyze recommendation sensitivity using Jacobian matrices.
- Study internal amplification behavior using Singular Value Decomposition.
- Detect abnormal directional amplification inside recommendation systems.
- Identify hidden manipulation and recommendation bias.
- Develop a model-level manipulation detection framework independent of behavioral analysis.
- Improve recommendation transparency and robustness.

4 LITERATURE REVIEW

4.1 Manipulation in Recommendation Systems

Modern recommender systems can exhibit hidden manipulation while still maintaining high evaluation metrics. Previous studies demonstrated that recommendation performance gains may hide manipulative recommendation behavior.

4.2 Web Injection and Behavioral Attacks

Researchers analyzed manipulation through web injection attacks and behavioral vulnerabilities. These approaches mainly focus on interaction-level attacks rather than internal model sensitivity.

4.3 Spectral Analysis for Anomaly Detection

Spectral signature analysis has been used for detecting poisoned data and backdoor attacks in machine learning systems. However, its application to recommendation systems remains limited.

4.4 Sponsored Recommendation Bias

Several statistical approaches have been proposed to detect sponsored recommendation bias. These methods focus primarily on promotional influence rather than broader spectral distortions.

4.5 Reinforcement Learning Manipulation

Research has shown that reinforcement-learning-based recommenders can manipulate user preferences using adaptive optimization strategies. These approaches are limited to specific architectures.

4.6 Shilling Attack Detection

Shilling attack detection methods analyze fake rating behavior and temporal interaction patterns. However, they do not investigate the internal mathematical behavior of recommendation models.

5 RESEARCH GAPS

The literature survey reveals multiple important gaps:

- Existing approaches mainly analyze user behavior rather than internal model dynamics.
- Jacobian sensitivity analysis is rarely explored in recommender systems.
- Spectral techniques such as SVD are underutilized for manipulation detection.
- Traditional metrics cannot reveal hidden amplification behavior.
- Internal recommendation sensitivity remains poorly studied.

These gaps motivate the development of a mathematical framework capable of detecting manipulation using internal sensitivity analysis.

6 METHODOLOGY

The proposed framework follows a multi-stage manipulation detection pipeline.

6.1 Input Processing

User interaction data is collected as input features.

Inputs may include:

- ratings
- likes
- clicks
- watch history
- browsing behavior
- purchase interactions

The recommender system processes these inputs to generate recommendation outputs.

6.2 Recommendation Function

The recommender system is mathematically represented as:

$$F(x) = y$$

where:

- x = input feature vector
- y = recommendation output vector

This representation allows mathematical analysis of recommendation sensitivity.

6.3 Jacobian Matrix Computation

The Jacobian matrix measures output sensitivity with respect to input perturbations.

$$J = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

Each element:

$$\frac{\partial y_i}{\partial x_j}$$

measures how much recommendation output y_i changes when input feature x_j changes slightly.

Large derivatives indicate high sensitivity and possible manipulation.

6.4 Singular Value Decomposition

Singular Value Decomposition is applied to the Jacobian matrix:

$$J = U\Sigma V^T$$

where:

- U = output sensitivity directions
- Σ = singular values
- V^T = input sensitivity directions

The singular values reveal dominant amplification behavior inside the recommender system.

6.5 Manipulation Detection

Manipulated systems often exhibit:

- unusually large dominant singular values
- concentrated sensitivity directions
- abnormal spectral distortion
- excessive recommendation amplification

These properties are used to identify hidden recommendation manipulation.

7 JACOBIAN AND SVD BASED MANIPULATION DETECTION

7.1 Recommender System as a Mathematical Function

A recommender system can be viewed mathematically as:

$$F(x) = y$$

where:

- x represents user input features
- y represents recommendation outputs

Inputs may include:

- user ratings
- clicks
- watch history
- likes
- interaction patterns

Outputs are recommendation scores generated for movies, products, or content.

7.2 Understanding Manipulation

Manipulation occurs when very small artificial input changes produce disproportionately large recommendation changes.

Example:
A small group of fake users gives extremely high ratings to a particular movie.
As a result:

- the movie suddenly appears in many recommendations
- recommendation rankings become distorted
- normal user preferences are overridden

This behavior indicates abnormal recommendation sensitivity.

7.3 Role of the Jacobian Matrix

The Jacobian matrix measures how sensitive recommendation outputs are to changes in input features.

Suppose:

$$x = [x_1, x_2, x_3, ...]$$

where:

- x_1 = rating for Movie A
- x_2 = rating for Movie B

Outputs are:

$$y = [y_1, y_2, y_3, ...]$$

where:

- y_1 = recommendation score for Movie 1
- y_2 = recommendation score for Movie 2

The Jacobian becomes:

$$J = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \cdots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \cdots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}$$

Each element:

$$\frac{\partial y_i}{\partial x_j}$$

means:

> "If input feature x_j changes slightly, how much does recommendation output y_i change?"

7.4 Example of Sensitivity

Suppose:

$$\frac{\partial y_5}{\partial x_2} = 0.1$$

This indicates that input feature x_2 has very little influence.

Now suppose:

$$\frac{\partial y_5}{\partial x_9} = 20$$

This indicates extremely high sensitivity.
Therefore:

- feature x_9 has abnormal influence
- recommendation outputs depend heavily on that feature
- manipulation may be occurring

7.5 Example Recommender System

Consider a simplified recommendation function:

$$F(x) = \begin{bmatrix} 5x_1 + x_2 \\ 5x_1 - x_2 \end{bmatrix}$$

where:

- x_1 = suspicious fake-rating feature
- x_2 = normal user feature

7.6 Computing the Jacobian

The Jacobian matrix becomes:

$$J = \begin{bmatrix} 5 & 1 \\ 5 & -1 \end{bmatrix}$$

Observe the columns carefully.
First column:

$$\begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

Second column:

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The first feature x_1 influences recommendation outputs much more strongly than x_2.
This implies:

- the system is highly sensitive to x_1
- fake ratings through x_1 can manipulate recommendations
- abnormal amplification is occurring

7.7 Applying Singular Value Decomposition

The Jacobian matrix is decomposed as:

$$J = U\Sigma V^T$$

SVD identifies:

- dominant amplification directions
- strongest sensitivity patterns
- hidden manipulation structures
- concentrated recommendation influence

7.8 Detecting Manipulation Using Singular Values

Normal recommender systems show smoothly decaying singular values:

$$[100, 80, 60, 40, 20]$$

Manipulated systems may show:

$$[500, 20, 10, 5, 2]$$

One dominant singular value becomes excessively large, indicating abnormal recommendation amplification.

8 IMPLEMENTATION DETAILS

The implementation focuses on mathematical sensitivity analysis and spectral distortion evaluation.

The framework performs:

- recommendation generation
- Jacobian computation
- spectral decomposition

- amplification analysis
- manipulation detection

The architecture can be extended to:

- collaborative filtering recommenders
- neural recommendation systems
- reinforcement learning recommenders
- hybrid recommendation architectures

The framework also supports future integration with real-time monitoring systems.

9 RESULTS AND OBSERVATIONS

9.1 Manipulation Detection Scores

The proposed system successfully distinguished normal recommendation behavior from manipulated recommendation behavior.

Detector Scores

- Normal Recommendation Score: 1.838
- Manipulated Recommendation Score: 4.194

Manipulated systems exhibited significantly larger spectral distortion.

9.2 Spectral Analysis Observations

The manipulated recommendation system showed:

- stronger dominant singular values
- concentrated sensitivity directions
- abnormal recommendation amplification
- unstable recommendation behavior

9.3 Recommendation Analysis

The detector identified suspicious recommendation amplification among highly promoted movies such as:

- Heat
- Star Wars Episode IV
- Seven
- Up (2009)
- The Wolf of Wall Street (2013)

10 CHALLENGES FACED

Several technical challenges were encountered:

- computing stable Jacobian matrices for large datasets
- handling high-dimensional recommendation spaces
- avoiding numerical instability during SVD
- distinguishing natural popularity from manipulation
- selecting reliable manipulation thresholds
- scaling the framework for real-world recommendation systems

11 APPLICATIONS

The proposed framework has multiple practical applications:

- manipulation detection in e-commerce recommenders
- fairness monitoring in OTT platforms
- sponsored recommendation detection
- fake rating amplification detection
- recommendation transparency analysis
- robustness evaluation for AI recommendation systems
- shilling attack identification

12 FUTURE SCOPE

Future improvements may include:

- real-time recommendation monitoring
- adaptive spectral anomaly detection
- deep-learning-based manipulation analysis
- integration with fairness-aware recommendation models
- large-scale deployment in commercial platforms
- hybrid sensitivity and behavioral analysis systems

13 CONCLUSION

This project proposed a Jacobian spectral distortion framework for detecting manipulation in recommender systems.

The recommender system was mathematically modeled as a transformation from user input space to recommendation output space. Jacobian sensitivity analysis was used to study how recommendation outputs respond to small input perturbations. Singular Value Decomposition was then applied to identify dominant amplification directions and hidden spectral distortions.

The study demonstrated that manipulated recommendation systems exhibit:

- abnormal amplification behavior
- concentrated sensitivity directions
- dominant singular values
- unstable recommendation dynamics

Unlike traditional behavioral approaches, the proposed framework provides a model-level manipulation detection mechanism based on internal mathematical behavior.

The work establishes the effectiveness of Jacobian and SVD-based sensitivity analysis for improving fairness, transparency, and robustness in recommendation systems.

14 REFERENCES

1. Z. Zhu et al., “Understanding or Manipulation: Rethinking Online Performance Gains of Modern Recommender Systems,” ACM RecSys, 2024.

2. Y. Zhang et al., “Understanding the Manipulation on Recommender Systems through Web Injection,” IEEE TIFS, 2020.

3. B. Tran, J. Li, and A. Madry, “Spectral Signatures in Backdoor Attacks,” NeurIPS, 2018.

4. R. Sen et al., “Detecting Sponsored Recommendations,” ACM EC, 2015.

5. M. Sparr et al., “Explicit User Manipulation in RL Recommender Systems,” FAccT, 2022–2023.

6. W. Zhou et al., “Shilling Attack Detection for Recommender Systems,” PLOS ONE, 2018.